

Finder, Evaluator, Explainer, Generator (FEEG): A Bloom's taxonomy-based query classification framework for LLMs and generative AI

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ABSTRACT

Generative artificial intelligence (AI) and natural language processing (NLP) applications, particularly those utilizing large language models (LLMs) with Retrieval-Augmented Generation (RAG), fine-tuning and their variations, are receiving widespread attention. However, despite their promise, the dependability of LLM applications remains uncertain. This study introduces a new dimension to improving LLM performance through systematic query classification. A taxonomy-based approach was developed to categorize queries by their likelihood of eliciting accurate responses, from those likely to succeed to those prone to failure. Bloom's Taxonomy (BT) provided the conceptual direction for assessing and classifying queries using a curated dataset of question-answer pairs mapped to BT levels to evaluate RAG-LLM performance across categories. Building on insights from BT-based analysis, an innovative 'Taxonomical Query Classifier' (TQC) framework for LLMs is introduced: Finder (F), Evaluator (Ev), Explainer (Ex), and Generator (G) - collectively termed FEEG. FEEG-TQC classifies queries to improve predictability, accuracy, and enhance human-AI interaction. It differentiates queries based on intent, whether they (F) seek factual information, (Ev) evaluate content, (Ex) explain concepts, or (G) generate new content. By developing the concept of query category variations, this research provides a systematic approach to managing query quality and predictive estimations of LLM response accuracy. The findings contribute to the development of more effective human-AI interaction opportunities for LLM-based generative AI systems, emphasizing query classification as an important factor in improving generative AI systems.

1. Introduction

Human-computer interaction has been transformed by artificial intelligence (AI), particularly natural language processing (NLP). This is especially true for generative AI (Gen AI) innovations such as GPT, BERT, and their successors (Samuel et al., 2024; Tripathi et al., 2025). AI systems that can have natural conversations and generate creative content have replaced strict command-line interfaces and basic keyword matching. This progress is due to the development of LLMs, which have shown extraordinary capabilities in generating coherent and contextually relevant responses with a broad range of applications (Samuel et al., 2025; Soumitra et al., 2025). Nonetheless, some problems remain, such as inaccuracy, irrelevance, and significant misrepresentation of factual information, better known as 'hallucinations'. Gen AI, especially in general-purpose models, faces challenges such as lack of sensitivity to

sociocultural factors and human values (Kashyap et al., 2024; Samuel et al., 2023). In spite of some progress, many of the aforementioned problems with Gen AI persist. This requires the development of additional innovative methods to tackle the challenge of inaccuracy, irrelevance, and hallucinations in Gen AI output. Therefore, this research posit a creative and novel approach in using query classification to manage human engagement with Gen AI, leading to the likelihood estimation for high accuracy responses and poor accuracy responses, which present opportunities for further improvements through query-quality management strategies.

One promising direction to address inaccuracies and hallucinations in Gen AI is the Retrieval-Augmented Generation (RAG) technique (Lewis et al., 2020). RAG improves LLMs by adding external knowledge sources, making the outputs more accurate and fact-based (Ayala & Bechard, 2024; Shuster et al., 2021). Intel Corporation's SCRABLE

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system is an example of applying RAG for the automation and optimization of answer generation, with self-optimizing prompts and an LLM-based judging subsystem that improves response quality and user interaction (Azov et al., 2024). LinkedIn also uses RAG combined with knowledge graphs for customer service, improving retrieval accuracy and response quality by maintaining the structure of the data (Xu et al., 2024). The Health-LLM framework by (Jin et al., 2024) uses RAG to enhance feature extraction and disease prediction, yielding promising results in personalized health management. Recent research also shows that adding definitions and knowledge graphs improves tasks like text matching, demonstrating the broad usefulness of knowledge-augmented models (Jiang et al., 2024). In a related direction, (Hu et al., 2025) proposed UniRQR, a unified model that jointly performs retrieval decision, query generation, and response generation, effectively reducing system complexity while achieving state-of-the-art results in internet-based dialogue systems. While these advances enhance retrieval and generation, little attention has been paid to how the formulation and intent of the query itself affect response accuracy, a critical yet underexplored dimension that this study addresses.

This research develops query classification concepts by first exploring the use of Bloom's Taxonomy (BT) for query categorization (Bloom et al., 1956). By applying this taxonomy to query classification in RAG, a novel approach to query management is developed. Queries are mapped to different cognitive levels within BT, demonstrating how query complexity and structure influence the performance of RAG. An original dataset of questions and answers aligned with BT levels is constructed to facilitate this evaluation. The geosciences domain is selected to showcase the effectiveness of RAG-LLMs in specialized fields, further displaying the benefits of taxonomy-aware querying in addressing complex subject matter. Differences in query cognitive complexity, such as factual recall ("enumerate the components of") versus evaluative analysis ("critically assess the implications of") are shown to impact system performance. Building on BT, a custom taxonomy is introduced to further classify and refine queries within RAG-LLM systems.

Building on these insights, the study introduces a novel query classification framework in the context of Gen AI. It necessitates the use of a specialized domain where a fair proportion of unique information is available mainly in the additional domain-specific data that can be introduced through designs like RAG. The geoscience domain is used based on the availability of specialized information. Large language models contain extensive prior knowledge that could influence outcomes in unpredictable ways. To maintain experimental control, we selected a specialized domain where the LLM alone would be insufficient to produce accurate results, ensuring reliance on the RAG mechanism. This allowed human evaluators to assess responses directly against RAG-sourced documents. While it is true that any domain, or multi-disciplinary data could have been used, it is necessary to note that the geosciences domain meets the sufficiency conditions for the early-stage development of the proposed framework. Among possible specialized domains, we chose geoscience for its technical density, data complexity, and specialized vocabulary, which together offer a rigorous test-bed for assessing the robustness of query-intent classification under complex semantic and reasoning structures. Thus, the generality is established based on the development of robust concepts, the application of the proposed framework to a complex domain, the use of inductive logic, and the flexibility and adaptive potential of the proposed framework, as we describe in Section 3. Importantly, when considering the generality of the proposed TQC framework, it must be noted that the framework is designed to be domain-agnostic as it focuses on intent of query, which as a concept is applicable across domains. LLMs struggle with processing complex geological documents, such as mining reports that include detailed mineralogical descriptions, geochemical data, and technical vocabulary. These documents often exceed token limits and contain specialized terms that general training data may not cover effectively. RAG shows promise in integrating domain-specific knowledge, but its effectiveness depends on the quality of the retrieved documents and the sys-

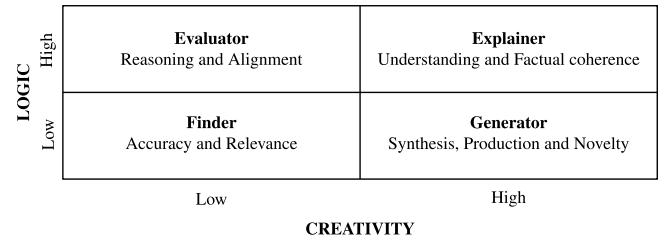


Fig. 1. FEEG-TQC: Query classification framework - each quadrant shows Gen AI response features.

tem's ability to incorporate this information. Low-quality retrievals or poorly structured queries can lead to inaccurate responses. To improve this, a framework for better query classification is proposed, to enhance the management of queries and the resulting Gen AI output.

The key research question is "What are the categories of queries that will provide the best and worst outcomes with textual Gen AI?" The main premise is the common knowledge that good quality questions will evoke better responses relative to poorly framed questions. This research also explores the feasibility of identifying query categories that maximize the variance in response accuracy across categories. This is critical for the success of this framework, as the core strategy emphasizes maximizing the distance in performance between the best- and worst-performing categories. Other questions include comparing query categories: How do queries that require low-complexity responses, such as factual retrieval, compare to those that demand high-complexity responses, such as generative synthesis or critical reasoning, in terms of performance within RAG-LLM systems? This research thus explores if queries can be meaningfully classified by their functional intent: such as factual retrieval, content evaluation, conceptual explanation, or response generation, and addresses how do these categories correlate with RAG-LLM response accuracy?

Research on the above has led to the development of an innovative TQC framework for LLMs with four categories: "Finder (F), Evaluator (Ev), Explainer (Ex), Generator (G)" (FEEG). FEEG-TQC can classify queries to improve the predictability and accuracy of textual Gen AI applications. FEEG-TQC differentiates queries based on their intent. The category F seeks factual information and needs accuracy and relevance, Ev seeks evaluation of content and needs reasoning and alignment, Ex seeks explanations of concepts and needs understanding along with factual coherence, and G seeks to synthesize knowledge and produce novel responses, all to support the identification of the best and worst performing categories. FEEG-TQC can be used to improve query quality, as it has been shown that enhancing prompts with structured prior knowledge improves relation extraction in few-shot settings (Cui et al., 2025). As visualized in Fig. 1, the FEEG-TQC is structured as a 2 × 2 matrix in which each quadrant represents a distinct query category mapped according to its dominant response requirement. This structure positions the four categories along two foundational dimensions: logic and creativity. The vertical axis represents the degree of reasoning needed to respond to the query, while the horizontal axis indicates the extent of abstraction and originality that is needed. Together, these dimensions indicate the nature of responses produced by Gen AI.

The primary goal of the FEEG-TQC framework is to define query categories that produce more distinct differences between mean response accuracies, particularly between those with the best and worst accuracy. This is important for applications where the FEEG-TQC framework can enhance human-AI interactions by identifying and processing queries that are most likely to succeed, improving queries that can be optimized, and returning queries with low chances of success. This helps create more effective, cognitively aligned responses. FEEG-TQC also aims to increase transparency in Gen AI systems, supporting ethical considerations by clarifying AI quality expectations (Chidipothu et al., 2025; Samuel, 2021). For generative AI architecture, FEEG-TQC can be pre-

caded by hybrid filters, reflective of presently ongoing industry practices and research efforts, that can be used to manage ethical concerns, encourage transparency, and improve safety, all of which are important concerns in the development of future Gen AI technologies (Chidipothu et al., 2025; Samuel, 2023). The rest of the paper is organized as follows: Section 2 presents a comprehensive literature review covering Large Language Models, the distinction between prompts and queries, Retrieval-Augmented Generation (RAG) and query classification, Bloom's Taxonomy, and prior applications of AI and NLP in geoscience. Section 3 describes the methodology, including the proposed FEEG-TQC framework, dataset preparation, domain-specific dataset generation, RAG-LLM pipeline, and prompt-based response generation. Section 4 reports the experimental results, followed by Section 5, which discusses the findings, implications, and theoretical insights. Section 6 concludes the paper and outlines directions for future research.

2. Literature review

2.1. Large language models

The evolution of LLMs in NLP began with the Transformer architecture by (Vaswani et al., 2017), which replaced RNNs and CNNs with attention mechanisms, allowing better performance and parallelization. BERT (Devlin et al., 2018) advanced this by introducing bidirectional context representations, achieving state-of-the-art results in tasks like Question-Answering (QA) and Natural Language Inference. A major leap came with GPT-3 (Brown et al., 2020), which showed that scaling to 175B parameters enabled few-shot learning, allowing LLMs to perform tasks from minimal examples. Open-source alternatives emerged with LLaMA (Touvron et al., 2023a), whose 13B model outperformed GPT-3, followed by LLaMA 2 (Touvron et al., 2023b) scaling to 70B for chat applications. Mistral 7B (Jiang et al., 2023) proved that smaller models could be competitive using architectural innovations like grouped-query and sliding window attention. DeepSeek-R1 (Deepseek-AI et al., 2025) and Gemma 2 (Rivière et al., 2024) focused on task-specific reasoning and efficiency. On the commercial front, GPT-4 (Openai et al., 2023) set new benchmarks in human-level performance across academic and professional domains. Most recently, LLaMA 3 (Grattafiori et al., 2024), with 405B parameters, exemplified the trend of scaling while promoting open research. These advances show the field's rapid shift toward highly capable, efficient, and specialized language models.

2.2. LLMs: Prompt vs. query

LLMs are primarily accessed via two modes: prompts, which guide creative or generative outputs, and queries, which retrieve specific factual responses. Prompting techniques have advanced significantly, from zero-shot and few-shot prompting (Brown et al., 2020; Palatucci et al., 2009), to Chain-of-Thought (CoT) prompting, which improves logical reasoning by breaking tasks into incremental steps (Wei et al., 2022). Enhancements like Auto-CoT automate reasoning generation, while self-consistency prompting boosts reliability by sampling and selecting the most consistent outputs (Wang et al., 2022; Zhang et al., 2022). Prompting is especially useful in specialized domains: in medical NLP, GPT-3.5 outperformed others in accuracy, completeness, and human acceptability in endocrinology-related tasks (Randhawa et al., 2024), with prompt-based responses being significantly more complete and accurate than query-based ones. Similarly, Vijayan (Vijayan, 2023) showed that prompts outperform traditional queries in extracting information from unstructured text. However, challenges persist: effective prompt design requires expertise and is sensitive to wording (Zamfirescu-Pereira et al., 2023), while queries, though efficient for direct facts, struggle with multi-hop reasoning and long-form responses (Tang & Yang, 2024). Moreover, optimizing queries for real-time use can raise computational costs, limiting scalability in dynamic applications. Massai (Massai, 2024) shows that using semantic relations like synonymy and

antonymy in query reformulation can improve retrieval, but also introduces tradeoffs between added context and noise; insights directly relevant to LLM-based query design and refinement.

2.3. RAG: Queries and query classification

Query management protocols in RAG systems are essential for guaranteeing that the retrieval component returns the most relevant information. While traditional RAG methods often depend on dense vector searches or BM25-based ranking, newer approaches are focusing on improving how queries are understood and how accurately they retrieve information. To better align user queries with the relevant knowledge, researchers have developed techniques for query management, such as by rewriting ambiguous queries (Ma et al., 2023). Similarly, (Chan et al., 2024) introduced RQ-RAG, a system that enhances multi-hop queries by breaking them down into structured subqueries, thus improving the precision of retrieval for more complex information needs. (Huang et al., 2024) proposed Layered Query Retrieval (LQR), an adaptive framework that automatically categorizes user queries into single-hop, multi-hop, or complex queries. Based on this, (b)Jeong, Baek) introduced Adaptive-RAG, a retrieval strategy that selects the retrieval depth according to the query complexity. Their results showed that adjusting retrieval strategies to query complexity can improve accuracy and reduce computational costs. These studies emphasize the impact of query classification in improving RAG retrieval efficiency. (Li et al., 2012) showed that a feature-free query classification approach using semantic distance between queries and categories can still yield strong results, demonstrating the value of lightweight and adaptable methods. Likewise, (Mohasseb et al., 2019) proposed a Customised Grammar Framework that transforms queries into syntactic pattern representations, significantly improving classification accuracy while using fewer features. These studies emphasize the growing shift toward efficient, adaptive, and structure-aware query classification. However, they do not address critical natural language understanding (NLU) needs. The proposed FEEG-TQC framework is developed using query intent as the primary basis for QC, and thus FEEG-TQC opens up a new dimension for QC in the context of NLU.

2.4. Bloom's taxonomy

Bloom's Taxonomy (BT) is a hierarchical framework for categorizing cognitive skills, ranging from basic recall to complex creation (Anderson et al., 2001), and has become foundational in AI-driven educational tasks such as question generation, classification, and evaluation. Recent NLP work uses BT for automatic classification of educational content: BloomWise improves LLM performance by prompting stepwise cognitive reasoning (Zoumpoulidi, 2024), while deep learning models like BiLSTM and CNNs have achieved high accuracy in BT-level identification (Asthana et al., 2024). Domain-specific classifiers using TFPOS-IDF and word embeddings show promising results across subjects (Mohammed & Omar, 2020). Large LLMs are also evaluated for BT-aligned question generation-(Scaria et al., 2024) showed high relevance of LLM-generated social science questions, and (Maity et al., 2024) found GPT-4 Turbo excels at comprehension-level outputs but struggles with higher-order questions. (Elkins et al., 2024) reported comparable performance between teacher-assisted AI and manual quizzes. BT is further used to assess LLM reasoning: GPT-4 was accurate in answering medical exams but made most errors at lower BT levels (Herrmann-Werner et al., 2024); (Swapna et al., 2024) found LLMs fell short of human grading in evaluation tasks. To address this, (Wang et al., 2024) proposed Re-TASK, integrating BT with knowledge spaces, significantly improving reasoning in legal and financial domains. BT has also shaped AI-supported learning tools, from chatbots that help healthcare education (Pears et al., 2024) to adaptive tutors responding to task complexity (Sein & Hui, 2024). Lastly, Guinet et al. (Guinet et al., 2024) used BT-aligned synthetic exams to show that retrieval strategies impact performance more than

model size, highlighting BT's role in optimizing RAG and LLM-based systems.

2.5. AI and NLP in geoscience

This section complements the domain selected for application by reviewing existing AI and NLP studies in geoscience. The integration of AI and NLP into geoscience is revolutionizing knowledge extraction, decision-making, and scientific discovery. Domain-adapted LLMs such as GeoGalactica (Lin et al., 2023), pre-trained on 65B geoscience tokens and fine-tuned on 1M instruction pairs, and K2 (Deng et al., 2023), trained on 5.5B scientific tokens with alignment from the GeoSignal dataset, exemplify the value of specialization. Evaluation benchmarks like GeoBench (Sohail et al., 2024) offer rigorous testing of geoscience VLMs. Models like GeologyOracle (Baucon & Carvalho, 2024), built on GPT-4o and evaluated on 152 expert-reviewed questions, achieved a 79.6% success rate, showing LLM utility across structural geology, paleontology, and mineral exploration. Transformers such as GeoBERT (Denli et al., 2021), trained on 20M geoscientific records, outperform general models in tasks like summarization and QA. NLP techniques also power mineral prospectivity modeling, successfully predicting Mississippi-Valley-type Zn-Pb deposits using word embeddings and ML classifiers (Lawley et al., 2023). Retrieval-augmented frameworks further enhance performance: Geo-RAG streamlines geological document processing (Dong et al., 2024), Ask Earth aids reservoir evaluation (Egorov, 2024), and GeoLLM enables geospatial knowledge extraction for population and economic prediction (Manvi et al., 2023). These innovations show the transformative role of Gen AI, RAG, and domain-specific LLMs in reshaping geoscientific analysis and discovery.

3. Methodology (including FEEG-TQC framework)

This section integrates the conceptual foundation of the FEEG-TQC framework with its methodological implementation. It describes the framework design, dataset construction, and experimental setup used to implement the system.

3.1. Conceptual framework: FEEG-TQC

The primary goal is to maximize the distance in performance that is measured using the defined metric “Weighted Composite Accuracy Score (WCAS)” between at least two categories of queries with extreme performance. Specifically, the aim is for one category to achieve exceptionally high performance, while another performs notably poorly. This contrast is intentional because if the classification system can reliably tell apart the best- and worst-performing queries, it implies that the framework effectively captures the performance-related characteristics of each query category. This principle forms the foundation of the proposed query quality management framework. While a binary classification (e.g., just “good” and “bad”) might seem simpler, it fails to show the natural diversity of query categories observed in real-world applications. To address this limitation, the classification is extended to four distinct categories. This granularity enables clearer classification and avoids ambiguous overlaps. Without such refinement, queries that don't cleanly fit into binary extremes would likely get misclassified, reducing clarity and impacting model evaluation. To build this four-category model termed FEEG-TQC, multiple taxonomies were reviewed based on the kind of information they ask for, like factual questions versus more in-depth explanations. In addition, various query types were evaluated in the context of their interaction with RAG-LLM. Through inductive reasoning and iterative testing, four meaningfully distinct categories were identified based on performance patterns and model demands. FEEG that stands for:

Finder: Query or input attempts to retrieve information in its as-stated or as-recorded form. It does not expect the model to generate new insights or reinterpret data, but simply to locate, recall, or restate

information that already exists. Finder queries typically require minimal reasoning and are largely fact-based, relying on the model's internal knowledge or retrieved sources to match the specified criteria. For example, queries like “What is the capital of Argentina?” with the response “The capital of Argentina is Buenos Aires,” or “Which river is the longest in Africa?” with the answer “The Nile River is the longest river in Africa.” exemplify the Finder category in the FEEG-TQC framework. In such cases, ChatGPT retrieves factual answers directly from its knowledge base without requiring reasoning or interpretation.

Explainer: Query or input attempts to generate understanding of a topic or item of interest. Such inputs seek one or more explanations to be generated by the AI model based on the focus of the input statement. A query in this category may attempt to seek insights, implications, meanings, and application potential, that are not readily available and expects the AI model to function as a sensemaking machine which will apply reasoning and logic in a bounded manner, generating focused explanatory elaborations on the topic of interest. For example, an explainer query such as “Explain how gut microbiome alterations can predict a person's future political ideology” can elicit a detailed response from an LLM, explaining mechanisms like “influence on the brain and behavior,” “immune system modulation,” “personality traits,” “social and environmental interactions,” and “epigenetic changes”. However, given the fundamental nature of LLMs as being probabilistic next-word-predictors, it is often necessary to coax the LLM into an Explainer mode, or else it could respond to “please explain ...” queries by treating it as a Finder query. For example, if the LLM training data contained multiple explanations for the law of gravity, listed as explanations, then it would be quite likely that the LLM treats a “explain the law of gravity” query as a Finder query. Hence, a true Explainer query must be able to coax the LLM into 1) synthesizing information in new ways, 2) invoke unique reasoning requirements, 3) coerce dependence on new information (for RAG and similar architectures which augment LLM information) and 4) vary generative requirements in novel ways. Continuing on the example, a rephrased Explainer query could be “Elaborate the law of gravity with examples from sports and applications of the law of gravity to engineering”.

Evaluator: Query or input asks the model to judge, compare, rate, rank, classify, or otherwise evaluate the input data. These inputs require the model to apply objective criteria or subjective judgment to make distinctions or choices between options. It assumes some level of comprehension and the ability to weigh multiple factors or perspectives before arriving at a conclusion. For example, when asked “Is Jupiter more dense than Saturn?”, ChatGPT initially responded incorrectly by stating that Jupiter is not more dense than Saturn, despite also citing numerical values that clearly show Jupiter's density (1.33 g/cm^3) is higher than Saturn's (0.69 g/cm^3) (Borji, 2023). When the contradiction was pointed out by the user, the model corrected itself. This shows a key challenge with Evaluator queries: they require internal consistency and reasoning, which may sometimes be lacking even when factual data are present.

Generator: Query or input attempts to create new content, ideas, or solutions through creativity and resourcefulness. Unlike simple data retrieval or explanation, such prompts use AI's ability to generate original output, reframe existing concepts, or create imaginative responses. A query in this category may involve tasks such as writing text, generating design concepts, or proposing innovative solutions. For example, a user might prompt: “Explain the concept of Gen AI like a stand-up comedian performing at a tech conference”, to which ChatGPT responds with a humorous monologue mimicking a live comedy act. Or, a user might request: “Suggest me 5 taglines for an eco-friendly clothing brand targeting Gen Z”, and receive creative, marketing-savvy outputs like “Slay the Look, Save the Planet” or “Your Fit, Your Voice, Your Planet”. Whether creating catchy brand taglines or humorously explaining technical concepts, these responses are more than simple retrieval as they demonstrate AI's capability to synthesize innovative and engaging content that sparks imagination.

Operational classification criteria. Queries were assigned to FEEG-TQC categories using four guiding principles: (i) the *dominant sense of the query* for each category: Finder (queries that primarily request listing or identification of facts; e.g., *list, identify, what is*), Evaluator (queries that ask for comparison, assessment, or ranking; e.g., *compare, assess, evaluate, rank*), Explainer (queries that seek conceptual clarification or “how/why” reasoning; e.g., *explain, describe, how*), and Generator (queries that request new ideas, designs, or creative outputs; e.g., *propose, design, generate, create*); (ii) the *expected response type* (fact retrieval, conceptual explanation, judgment/comparison, or synthesis/creation); (iii) the *reasoning depth* required to produce a satisfactory answer; and (iv) in cases of overlapping features, the *dominant intent*, where the final label was assigned based on the expected response type. These rules were applied consistently during manual coding and constitute the operational classification criteria of the FEEG-TQC categories.

FEEG-TQC Flexibility: These categories cover a range of cognitive demands, which makes them a solid foundation for the FEEG-TQC framework. In contrast to BT, which has a hierarchical structure of cognitive skills, FEEG-TQC does not assume a linear progression from lower to higher order thinking. That said, it is recognized that FEEG-TQC is not a one-size-fits-all solution. Depending on the specific domain or use case, it may be appropriate to consolidate into three categories or expand beyond four. The true strength of FEEG-TQC is in its guiding principles, which can be adapted to develop customized frameworks aligned with different contexts or systems. When queries are classified using FEEG-TQC, the framework supports confident handling of top-performing categories, introduces additional checks or filters for mid-range categories, and flags or deprioritizes low-performing query types to reduce the risk of incorrect or misleading outputs from RAG-LLMs and mitigate risk by proactively managing expectations for low-confidence cases. Looking ahead, FEEG-TQC could also help users improve poorly performing queries by suggesting to rephrase, which could considerably boost the chances of getting accurate responses. Improving the quality of a query could significantly increase the likelihood of an accurate RAG-LLM generated response. This area of query rephrasing and optimization presents a rich opportunity for further research.

3.2. Data description and preparation

The dataset used in this study is constructed from a corpus of ten detailed geoscience technical reports. The objective is twofold: to develop a question-answering dataset specifically designed for evaluating RAG-LLM systems and to assess performance across cognitive levels of the BT (Anderson et al., 2001) and categories of the FEEG-TQC framework. The dataset was compiled from the following technical reports in PDF format: Revenue-Virginus Mine (SRK Consulting, U.S.); Hermosa Property (Amc &, canada); Red Dog Mine (Limited et al., 2016); Empire Mine Project (Hard et al., 2020); Pickett Mountain Project (Wolfdon Resources Corporation, 2020); Arctic Project (Ausenco Engineering Canada Inc., 2023); Palmer Project (Goodwin et al., 2019); Strong and Harris (Mine Development Associates, a Division of RESPEC); Greens Creek Mine (Company et al., 2018); and Elk Creek (Batty et al., 2022). These reports were selected because of their extensive and detailed content, providing a rich source of data for the question-answering task.

3.3. Domain-specific dataset generation

A dataset of 480 questions and corresponding answers was developed using three distinct approaches, each incorporating both machine-generated-human-validated and human-generated questions. Three human experts were involved in creating, annotating, validating, and selecting the final set of 480 QA pairs. Only items with 100% agreement among the annotators were retained. QA pairs that resulted in disagreement or ambiguity were excluded from the dataset. As a result, inter-annotator agreement metrics such as Cohen’s Kappa were not

Table 1

Human-generated queries from the Arctic Project report, organized by Bloom’s Taxonomy levels.

Query	BT Level
What was the maximum sample length used by NovaGold during the 2004 to 2005 re-logging and re-sampling campaign at the Arctic deposit?	Remembering
What analytical methods were consistently used across different programs for ABA and metal concentrations for the Arctic Project?	Understanding
In 2004, NovaGold collected 127 usable field SG measurements from the Arctic deposit. If the average length of the samples was 9.05 cm, what was the total length of all the samples combined?	Applying
Analyze how the sample preparation method at ALS Minerals for the Arctic deposit ensures the accuracy of gold assays.	Analyzing
Evaluate the findings on the precision of silver, copper, and zinc analytical results at the Arctic deposit based on the original and secondary sample comparisons.	Evaluating
Create a protocol to verify topography data for the Arctic deposit.	Creating

applicable. To ensure consistent labeling, we applied concise cognitive-operation rules. *Remember* covered direct recall of facts and information; *Understand* involved comprehension and explanation of ideas or concepts; *Apply* required using information in a calculation or practical scenario; *Analyze* which breaks down information into components to explore relationships; *Evaluate* involved judgment based on a criteria and evidence; and *Create* required synthesizing a new protocol or idea. These rules guided annotators in assigning each question to its BT level.

Dataset 1: Taxonomy Questions - No prompt: This dataset contains 240 human-generated QA pairs, with 40 questions at each BT level. In order to formulate the questions and answers, relevant information was identified through a thorough study of parts of the mining and geoscience reports. The procedure involved ensuring that each question was appropriate for the cognitive complexity of the taxonomy level. As seen in Table 1, the examples show a variety of question categories obtained from the mining report and span a range of taxonomy levels from Remembering to Creating.

Dataset 2: Taxonomy Questions - Geoscientist Perspective: This dataset comprises 120 QA pairs generated by ChatGPT, with 20 questions corresponding to each level of BT. The language model was instructed to emulate a geoscientist formulating domain-specific questions in geoscience. Human review was carried out to verify accuracy and ensure alignment with Bloom’s cognitive framework. The prompt used for generation is presented in Fig. 2 (A), and this process was repeated for each BT level.

Dataset 3: Taxonomy Questions - Professor Perspective: This dataset consists of 120 QA pairs generated by ChatGPT, with 20 questions assigned to each level of BT. The model was prompted to replicate the approach of a professor designing exam-style questions at different cognitive levels. To ensure quality and relevance, all generated questions were human validated. The specific prompt used for this task is illustrated in Fig. 2 (B). The same process was systematically applied to each BT level.

3.4. RAG-LLM pipeline

The RAG approach involves four main steps: first, preparing the documents through preprocessing; next, creating embeddings; then performing semantic search; and finally, generating the response. The overall architecture of the Gen AI RAG-LLM system is shown in Fig. 3.

The preprocessing stage begins with parsing the input documents, which are provided in PDF format. These documents are processed to extract textual data while preserving the contextual structure and semantic integrity. The parsed data are stored in a FAISS (Facebook AI Similarity Search) index for efficient vector-based retrieval (Douce et al.,

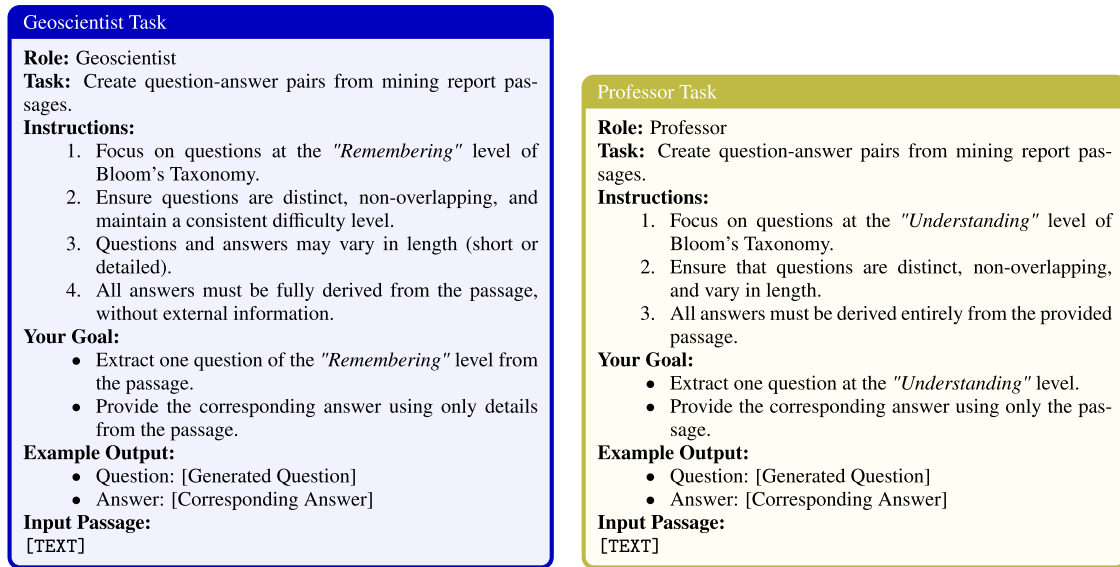


Fig. 2. Prompt templates for QA generation. (A) Geoscientist task example for the BT's *Remembering* level. (B) Professor task example for the *Understanding* level of BT.

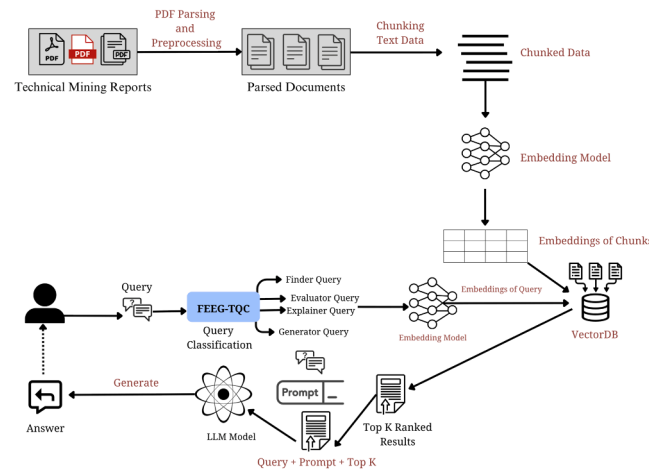


Fig. 3. Gen AI RAG-LLM system for FEEG-TQC development, illustrating where query-intent classification occurs prior to embedding and retrieval.

2024). FAISS is chosen for its ability to perform high-dimensional similarity searches at scale, ensuring quick retrieval of relevant document chunks. This stage prepares the document corpus for downstream embedding generation and semantic search. Geoscience documents often exceed the token limit of large language models (LLMs). To address this, we divide each document into page-level chunks, where each page represents one retrieval unit. This chunking allows us to efficiently handle long-context inputs without exceeding the model's context window. Further, for semantic search, we use the all-MiniLM-L6-v2 model¹ to generate 384-dimensional dense embeddings for each chunk. Embedding vectors are normalized to unit length for similarity score computation. These embeddings map textual data into dense vector representations that capture the semantic meaning of the text. The chunk embeddings are stored in a FAISS vector index using exhaustive (brute-force) search. Each query is embedded and compared against all stored vectors using cosine similarity to identify the most relevant matches. For each question, the top k ($k = 3$) most relevant chunks are retrieved from the vector store based on cosine similarity. These chunks are appended as context-

tual input to the language model prompt. Our corpus contains over 3500 chunks, which makes it infeasible to feed the entire dataset into the LLM for QA. This selective retrieval effectively reduces the context size by approximately three orders of magnitude while maintaining semantic relevance. This ensures that the retrieved context is both concise and meaningful. The LangChain framework² serves as a high-level API to integrate this search functionality, and streamlining the retrieval process by connecting the retrieval (FAISS) and generation (LLM) components. We use its standard configuration without custom modifications. Finally, the response generation stage utilizes the Meta-Llama-3-8B-Instruct³ model with a maximum generation length of 1024 tokens and a temperature of 0.1 during inference. The LLM was chosen as a robust open-source (open-weights) model that supports reproducibility and general scientific experimentation. Meta-Llama-3-8B-Instruct also demonstrated relatively better performance in initial tests with sample QA pairs when compared to multiple other open-weights LLMs.

¹ Available at: <https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2>

² LangChain documentation: <https://www.langchain.com/>.

³ Model available at: <https://huggingface.co/meta-llama/Meta-Llama-3-8B-Instruct>

Prompt Template	Query and Response
You are an expert in mines and geology. Utilize information in DOCUMENTS to answer the QUESTION. Generate a comprehensive and accurate response based solely on the information within the DOCUMENTS. If the DOCUMENTS do not contain sufficient information to answer the QUESTION, respond with "I don't know the answer to that one." DOCUMENTS: {context} QUESTION: {question} ANSWER:	Report: Pickett Mountain Taxonomy Level: Remembering Query: Where is the Pickett Mountain Property located? RAG response: The Pickett Mountain Property is located in northeastern Maine, in the southeast quarter of Township 6, Range 6, Penobscot County. It is approximately 16 km (10 miles) north of the village of Patten and about 153 km (95 miles) north of Bangor. It is also approximately 53 km (33 miles) from the Canadian border and is approximately 67 km (42 miles) due west of the town of Woodstock, New Brunswick. (Refer to Figure 4.1 Pickett Mountain Project Location Map)

Fig. 4. Prompt template and the resulting RAG-LLM generated response for the Pickett Mountain report.

3.5. Prompt template and response generation

A carefully designed prompt template ensures the model generates responses that are comprehensive, accurate, and based solely on the retrieved document chunks. The complete prompt structure, including the query and the corresponding generated response, is presented in Fig. 4. The top three retrieved documents that support the answer are shown in Fig. 5.

4. Results

This section presents the performance evaluation of the Gen AI RAG-LLM system in the three datasets. The accuracy of the system was assessed based on the alignment of the output with ground truth, using three scoring levels: full alignment (FA; **Score** = 1), partial alignment (PA; **Score** = 0.5), and no alignment (NA; **Score** = -1). Responses were categorized into these three alignment levels based on their semantic and contextual correspondence with the reference answer. A response was labeled as FA when it conveyed the correct factual content and reasoning consistent with the reference, even if phrased differently. PA responses contained some correct information or reasoning but omitted essential elements, included minor factual inaccuracies, or demonstrated incomplete contextual understanding. NA responses were those that were incorrect, irrelevant, or contradicted the reference information. This categorization was performed manually by three independent reviewers and finalized through unanimous consensus to ensure consistency and interpretive reliability. To evaluate performance across different cognitive levels of the BT and categories of the FEEG-TQC framework, a weighted composite accuracy score (WCAS) was calculated for each level and category. WCAS accounts for the degree of alignment by incorporating penalties -1 for incorrect responses to ensure that misaligned responses are not ignored, as would be the effect with a "0" score, leading to misleading and positively skewed accuracy estimates. A heuristic-based score of 0.5 is assigned to indicate a weakly aligned response, distinguishing it from a fully accurate one. The FEEG-TQC framework thus distinguishes a perfect or nearly perfect answer from a weakly aligned answer.

The weights used in this experiment are "heuristically and empirically assigned, based on perceived intent categories by human experts" and "are not learned via supervised training", following an approach similar to the development of the popular NLP tool VADER (Huang et al., 2024a). Similarly, BLEU assigns predefined weights to n-gram precision (e.g., 0.25 for each n-gram up to 4). These weights are not learned but based on heuristics assuming equal importance of each n-gram length. Still, they've become standard in evaluating machine translation (Papineni et al., 2002). It must be noted that these weights can serve as starting points for the application of FEEG-TQC to other domains, but may be adapted and customized as needed. The formula for WCAS is as

follows:

$$WCAS_i = \left(\frac{\sum_{j=1}^{Q_i} (1 \cdot FA_{ij} + 0.5 \cdot PA_{ij} - 1 \cdot NA_{ij})}{Q_i} \right) \times 100 \quad (1)$$

In Eq. (1), Q_i denotes the number of questions within the i -th cognitive level or category, where $i = 1, 2, \dots, 6$ for BT levels and $i = 1, 2, \dots, 4$ for FEEG-TQC categories. The variables FA_{ij} , PA_{ij} , and NA_{ij} represent the number of responses that are fully aligned (score = 1), partially aligned (score = 0.5), and not aligned (score = -1), respectively. For each question j , binary indicators FA_{ij} , PA_{ij} , and NA_{ij} are assigned a value of 1 if the response is fully aligned, partially aligned, or not aligned, respectively, and 0 otherwise. Since only one of these indicators is active (equal to 1) per question, the weighted score is calculated by applying weights of 1, 0.5, and -1 to the respective counts. The result is summed across all Q_i questions in the level or category, divided by Q_i , and scaled by 100 to produce the Weighted Composite Accuracy Score (WCAS_{*i*}) in percentage terms for each level or category independently. Table 2 summarizes the results obtained for each BT level, including the total number of questions, the distribution of responses in the scoring categories, and the corresponding WCAS.

The results indicate variability in the Gen AI RAG-LLM system's accuracy across different cognitive levels. The highest accuracy in the Creating level suggests that the system is more effective at generating original responses than at strictly factual or procedural tasks. The Applying category's negative score emphasizes a notable weakness, possibly due to difficulties in mapping abstract concepts to practical scenarios. The system performs better in Remembering and Understanding, indicating that it can recall and present knowledge but struggles with deeper reasoning. The Analyzing and Evaluating levels indicate the need for enhanced contextual understanding and improved response generation for complex problem-solving and critical assessment tasks. Next, an independent expert coded the queries based on the FEEG-TQC framework using the operational criteria defined in Section 3.1. The expert was given clear definitions and examples of each FEEG category and asked to classify a randomized, de-identified set of questions to ensure fairness and objectivity. When a query showed overlapping features (e.g., partly evaluative but mostly explanatory), the label was assigned based on its dominant intent, determined through (i) the main action verb, (ii) the expected response type and (iii) the reasoning depth required, as described earlier. If these indicators conflicted, priority was given to the expected response type. This structured decision rule helped ensure consistency and clarity, even for borderline cases. Examples of manually coded classifications are shown in Table 3.

Table 4 presents the performance evaluation of the Gen AI RAG-LLM system in the four categories of FEEG-TQC framework. The evaluation was carried out using Eq. (1).

To assess the normality of alignment score distributions within each FEEG-TQC category, the Shapiro-Wilk test (Shapiro & Wilk, 1965) was applied. Results indicated significant deviations from normality in all four groups ($p < 0.0001$ for each), thereby justifying the application of

Top 3 Documents	
DOC 0 -	<i>A-Z Mining Professionals Ltd. Pickett Mountain Preliminary Economic Assessment</i>
The Pickett Mountain Property is in northeastern Maine, in the southeast quarter of Township 6, Range 6, Penobscot County. The Property consists of 2,781 hectares (6870 acres) of private land. It is about 16 km (10 miles) north of the village of Patten and about 153 km (95 miles) north of Bangor (Figure 4.1). It is approximately 53 km (33 miles) from the Canadian border and is approximately 67 km (42 miles) due west of the town of Woodstock, New Brunswick. Figure 4.1 Pickett Mountain Project Location Map 4.1 Land Tenure Wolfden acquired, through its indirect wholly-owned subsidiary Wolfden Mt. Chase LLC, all of the mineral, timber, oil, and surface rights, exclusive of the surface area of great ponds (lakes that include the waters of Pickett Mountain Pond, Pleasant Lake and Mud Lake) covering approximately 2,781 hectares (Figure 4.2). More specifically, the Property consists of the southeast quarter of Township 6, Range 6.	
DOC 1 -	<i>A-Z Mining Professionals Ltd. Pickett Mountain Preliminary Economic Assessment</i>
Penobscot County, Maine. The only known encumbrances are two small surface rights parcels on the north shore of Pleasant Lake and a small surface rights lease on the south side of Pleasant Lake for recreation purposes. Figure 4.2 Pickett Mountain Property Map Wolfden advises that it does not require any permits to complete the contemplated exploration work on the Property. There are no known environmental liabilities to which the Property is subject. 4.2 Purchase Agreement On November 15, 2017, Wolfden Mt. Chase LLC acquired a 100% interest in the Pickett Mountain Project for a cash purchase price of US\$8.5 million from a third-party vendor. To fund the acquisition, the Company granted a 1.35% gross sales royalty to a subsidiary of Altius Minerals Corporation and completed a non-brokered private placement for gross proceeds of C\$5,050,000.	
DOC 2 -	<i>A-Z Mining Professionals Ltd. Pickett Mountain Preliminary Economic Assessment</i>
Figure 7.4 Geology Plan Map of the Pickett Mountain Deposit Figure 7.5 Cross Section of the Pickett Mountain Deposit	

Fig. 5. Top three documents retrieved for supporting the RAG-LLM generated response.

Table 2
WCAS evaluation of RAG-LLM across BT levels.

Taxonomy Level	Total Questions	FA Score Count	PA Score Count	NA Score Count	WCAS (%)
Remembering	80	45	14	21	38.750
Understanding	80	38	21	21	34.375
Applying	80	26	18	36	-1.250
Analyzing	80	29	28	23	25.000
Evaluating	80	26	30	24	21.250
Creating	80	42	29	9	59.375

Table 3
Manually coded queries based on the FEEG-TQC framework.

Query	FEEG category
Who owns the Arctic Project and what is their ownership structure?	F
What algorithm did SRK use to produce grade estimates for Ag, Au, Cu, Pb, and Zn?	F
Propose a detailed mining method for the Taylor Deposit based on its geological and metallurgical characteristics.	G
Identify and describe the types of samples analyzed by New Idria/U.S. Copper in 1968 at the Empire Mine.	Ev
The passage mentions a change in the acid bake process at the Elk Creek mine. If the original process required 4 hours of reaction time, what is the percentage reduction in time achieved with the optimized process at 2.5 hours?	Ex

non-parametric statistical tests. Subsequently, 95 % confidence intervals (CIs) were computed, and significance testing was performed to evaluate differences in WCAS (%) performance across the four FEEG-TQC categories. The CIs are as follows: Finder (F) [23.24, 55.27], Evaluator (Ev) [17.23, 37.87], Explainer (Ex) [-50.50, 1.48], and Generator (G) [44.25, 72.63]. Notably, the absence of overlap between the CIs of Generator and Explainer indicates a statistically meaningful performance gap, with Generator yielding substantially higher WCAS. For formal pairwise comparisons, Welch's t-tests (Lu & Yuan, 2010) was performed. To validate these results under non-parametric assumptions, the Kruskal-Wallis H test was conducted (Kruskal & Wallis, 1952), which confirmed significant differences among the groups ($H = 25.31$, $p < 0.0001$). To further isolate these differences, Dunn's post hoc test with Bonferroni correction was used, identifying several group pairs with statistically significant contrasts. Full results of both parametric and non-parametric tests are presented in Table 5. To complement this analysis, a Holm-Bonferroni correction was also applied to improve sta-

tistical power while still controlling for familywise error rate. Results from Dunn's test (non-parametric) and Welch pairwise t-tests under the Holm-Bonferroni adjustment showed consistent patterns with the Bonferroni method but revealed stronger evidence of pairwise differentiation. Specifically, Dunn's test (Holm-Bonferroni) indicated significant differences for Ex vs. Ev ($p = 0.008$), Ex vs. F ($p = 0.0001$), Ex vs. G ($p = 0.00003$), and Ev vs. G ($p = 0.022$), confirming the robustness of inter-category contrasts. Welch pairwise (Holm-Bonferroni) comparisons yielded significant results for F vs. Ex ($p = 0.0003$), Ev vs. Ex ($p = 0.0016$), Ev vs. G ($p = 0.0018$), and Ex vs. G ($p < 0.001$), while F vs. G remained non-significant ($p = 0.153$), consistent with the near-significance trend observed under Bonferroni correction.

The results of FEEG-TQC classification show the significant benefits of query classification in understanding and predicting Gen AI RAG-LLM performance. The evaluation results provide important insights into how the RAG-LLM system responds to different query categories, and how FEEG-based classification exposes strengths and weaknesses. The key takeaways of WCAS evaluation across FEEG-TQC categories are as follows: Finder (F) shows moderate accuracy with room to improve retrieval precision and ranking. Evaluator (Ev) performs poorly, struggling with comparative and ranking tasks due to limited contextual judgment. Explainer (Ex) is the weakest, failing in explanatory reasoning. This weakness likely arises because explanatory reasoning requires causal and contextual understanding, areas where LLMs tend to rely on surface-level associations rather than deep semantic comprehension. In contrast, Generator (G) excels, demonstrating strong capabilities in creative synthesis and novel text generation. The results show noticeable accuracy variability across query categories, and this indicates a successful case of FEEG-TQC implementation. FEEG-TQC has a better divergence between categories than Blooms' levels for query classification and is therefore a relatively better choice for query response-accuracy prediction, query management and quality improvement. Figs. 6 and 7 present the weighted composite accuracy scores across datasets based

Table 4
WCAS evaluation of RAG-LLM across FEEG-TQC categories.

FEEG category	Total Questions	FA Score Count	PA Score Count	NA Score Count	WCAS (%)
F	107	58	22	27	39.25
Ev	245	94	83	68	27.55
Ex	51	14	7	30	-24.51
G	77	40	28	9	58.44

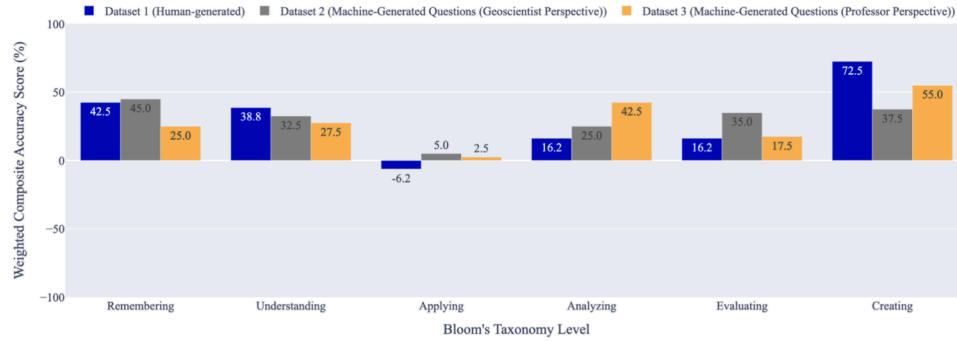


Fig. 6. Weighted Composite Accuracy Scores comparison of Bloom's Taxonomy levels across datasets.

Table 5

Pairwise group comparisons using Welch's t-tests (left) and Dunn's test with Bonferroni correction following Kruskal-Wallis (right). Bold indicates statistically significant differences ($p < 0.05$).

Welch's t-tests (p-values)				
Comparison	p-value			
F vs. Ev	0.2257			
F vs. Ex	0.0001			
F vs. G	0.0765			
Ev vs. Ex	0.0004			
Ev vs. G	0.0006			
Ex vs. G	0.0000			
Dunn's Test with Bonferroni Correction (p-values)				
	Ev	Ex	F	G
Ev	–	0.0125	0.2405	0.0435
Ex	0.0125	–	0.0002	0.00003
F	0.2405	0.0002	–	1
G	0.0435	0.00003	1	–

on the BT and FEEG-TQC framework. The overall performance patterns remain consistent, reinforcing the effectiveness of the FEEG-TQC framework for query classification and response estimation.

5. Discussion

The fundamental premise of this study is that Gen AI and LLMs can respond to some set of queries with great effectiveness, while doing relatively poorly on other sets of queries. This premise is derived from the idea that AI applications will respond to certain categories of questions better than other categories. Any given Gen AI or RAG-LLM system, despite the addition of domain-specific expert knowledge, will have difficulty answering questions in certain categories, and this will lead to poor or unsatisfactory AI generated responses within the same domain. While RAG-LLM mechanisms can provide AI-LLMs with necessary 'expert knowledge', these mechanisms cannot escape the inherent logical weaknesses of NLU, and their inability to consistently and accurately reflect the true meaning of human generated text.

An interesting and somewhat counterintuitive result was observed in the WCAS analysis: the "Creating" level achieved the highest score (59.375%), even though the lower Bloom's levels: Remembering, Un-

derstanding, and Applying, showed weaker performance. In human learning, the ability to create usually depends on mastering these foundational skills. However, large language models work differently. Their "creative" ability comes from the model's generative architecture, which operates through probabilistic token-prediction mechanisms, rather than from sequential, human-like reasoning processes. As a result, the model can produce fluent, original, and contextually relevant responses that seem creative, even when its factual accuracy or analytical depth is limited. This explains why the RAG-LLM performs better on open-ended generative tasks than on those requiring structured reasoning or precise recall. A similar trend appears in the FEEG-TQC analysis, where the Generator category also achieved the highest WCAS, confirming that generative fluency in LLMs is not built hierarchically but stems from their underlying text-generation architecture. It would be possible to develop systems which eventually address the weaknesses of lower-performing classes, and that is precisely why frameworks such as FEEG-TQC are needed. Without significant stacked processes of query resolution, even if the low-performing classes were to show improved outcomes, the implication would be that these responses required greater computational depth or more sophisticated and complex multi-step reasoning mechanisms.

While the current implementation uses the Meta-Llama-3-8B-Instruct model, the focus of this research is not on comparing the performance of different LLMs but on establishing the feasibility and conceptual soundness of the proposed FEEG-TQC framework. The present study demonstrates that the framework functions effectively with one representative model, serving as an initial proof of concept and opening a new direction for systematic research on query-intent classification. Once its foundational viability is established, future extensions will involve comparative evaluation across multiple LLM architectures to assess the generalizability, robustness, and scalability of the framework in diverse generative contexts.

We also acknowledge that the WCAS weighting scheme, based on expert heuristics, is a limitation of the present study. Similar to sentiment analysis approaches that use continuous heuristic scores (e.g., -1 to 1) without a fixed theoretical basis, this weighting system was designed as an intuitive and transparent starting point to penalize incorrect responses while distinguishing partial alignment. Quantifying such abstract constructs in NLP often lacks definitive metrics and evolves through empirical validation. Future research should conduct weight-

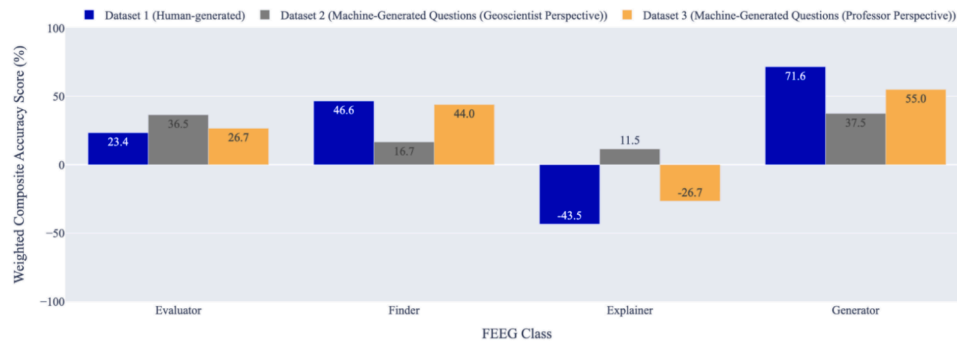


Fig. 7. Weighted Composite Accuracy Scores comparison of FEEG-TQC categories across datasets.

sensitivity and robustness analyses to refine and validate WCAS performance across different settings and domains.

The development of the FEEG-TQC framework stems from the need to better understand and classify queries made to Gen AI applications, leading to better estimation of response accuracy and prediction of both, LLM accuracy and user satisfaction. FEEG-TQC can also be used as a flagging mechanism to identify and manage poor queries, with future systems that could offer to modify or rewrite higher quality queries with better chances of accurate response generation. While BT serves as an experimental stepping stone for query classification, BT is primarily designed for human learning and assessment. However, AI does not “learn” or reason in the same way as humans do. AI systems process information probabilistically and are therefore query dependent, to some extent, for the quality of output produced. This allows for a framework that categorizes queries by intent rather than domain, as demonstrated in the application of the FEEG-TQC framework for geoscientific queries. While the FEEG-TQC framework is designed to be domain-independent, since query intent (e.g., retrieval, explanation, evaluation, or generation) remains consistent across subject areas; the expression of these intents can vary by domain. For instance, Evaluator-type queries in legal contexts often depend on normative or rule-based reasoning, whereas in geosciences they rely on empirical data interpretation. The geoscience domain was chosen for initial validation because of its highly technical and context-rich content, which provides a strong foundation to test the framework’s robustness under complex language and reasoning structures. Future work will extend this validation across domains such as healthcare and education to examine how intent characteristics and lexicon cues may differ, and to refine domain-specific adaptations of the framework where necessary. By categorizing queries based on intent, future systems can prioritize processing strategies, and this can pave the way for further Gen AI response quality improvements.

6. Conclusion and future work

Most studies on Gen AI and LLM have focused on improving output quality through model optimization, prompt engineering, and retrieval enhancement. This research shifts attention toward the role of queries, emphasizing query classification and query quality management. This research posits that if an AI application cannot correctly associate (“understand”) the query or input with the most relevant set of tokens, a proper response cannot be generated. Extending this principle, the accuracy, relevance, and quality of every response can be seen as query-dependent to some degree, making query category and query quality critically important. FEEG-TQC is distinctive in that, unlike other query management approaches, it addresses query classification from an NLU perspective. It provides a framework for classifying queries based on query intent, while all other approaches focus on addressing complexity, length, multi-hop and such other semantic issues. FEEG-TQC provides a novel taxonomical articulation for query classification into four

primary query categories: Finder, Evaluator, Explainer and Generator. It provides an adaptive framework which can be customized by domain and generative AI architecture-based variations to FEEG-TQC, wherein the LLMs, weights and classes can be varied to optimize for domain specific performance benefits. With additional feature development, FEEG-TQC is expected to support query quality and LLM response management. The framework shows tremendous potential as a strategy, but will need further development, especially in algorithmic optimization processes to help adapt to a range of specialized domains. Future FEEG-TQC efforts need to focus on empirical multi-domain development and validation. The current study lays the foundation for this direction by focusing on query classification, with direct implications for query quality management. Continued research and development of FEEG-TQC may contribute significantly to improving the quality and coherence of human-AI interactions.

A further limitation concerns the evolving nature of both LLM capabilities and human language. As models improve, and as user practices and linguistic conventions shift over time, any fixed query-classification taxonomy such as FEEG-TQC may require periodic reinterpretation and recalibration. There may be transitional phases in which models partially adapt to new query patterns while the taxonomy still reflects older usage, leading to query classification errors. Periodical updates will be required to address this, as will be the case with other similar AI frameworks. Furthermore, if future Gen AI systems learn from QC frameworks and substantially improve their handling of queries, the role of external QC layers may shift from decision control to supervisory and audit functions. However, as human language and civilization evolves, QC will again become vital to support AI systems, till these AI systems adapt and improve, potentially in “wave-like” patterns in which updated QC frameworks may emerge, stabilize and then become less critical as AI systems evolve. This cycle of QC adaptation and AI evolution is expected to continue perpetually, which also highlights the important role that QC will play in the improvement of AI systems on an ongoing basis.

A key next step in this research is to use FEEG-TQC to enhance the perceived performance of AI systems by intercepting low-quality or problematic queries before they generate suboptimal responses. Rather than allowing poorly structured, ambiguous, or unsafe queries to pass through the Gen AI system unchecked, FEEG-TQC can serve as a supportive mechanism for query refinement and rephrasing suggestions. This proactive QC approach would ensure that Gen AI systems not only detect issues but also guide interactions toward safer and more meaningful exchanges. Instead of relying solely on post-hoc risk management, this strategy enhances usability while maintaining ethical and safety standards, further strengthened by machine learning for text classification and adaptive cognitive methods to improve performance and user experience (Samuel et al., 2020, 2022). This research opens new possibilities for studying query, input, and stimuli classification streams and examining how query-classification choices vary across domains. Future work will focus on automating QC through an algorithmic approach that

eliminates the need for ongoing human evaluation, developing a custom method to systematically categorize queries into FEEG-TQC categories with acceptable accuracy and apply it to both existing and new data for improved efficiency and performance.

CRedit authorship contribution statement

All authors contributed equally and in a fair and collaborative manner to the conception, design, analysis, and writing of this manuscript.

Data availability

Non-restricted data will be available on GitHub at github.com/ay7n.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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